

YIHANG CHEN

+1 (619) 403-8586 · yihangchen2003@ucsd.edu · www.linkedin.com/in/cyh2003/

EDUCATION

UC San Diego, La Jolla, CA

Sep 2025 – (Expected) Dec 2026

M.S. in Computer Science and Engineering (CS75)

Zhejiang University, Hangzhou, China

Sep 2021 – Jun 2025

B.Eng. in Computer Science

Overall GPA: 3.97 / 4

Related Courses: Data Structure, Operating System, Database System, Theory of Computation, B/S Architecture Software Design, Artificial Intelligence, Natural Language Processing, Multimedia Technology

SKILLS

- **Machine Learning:** PyTorch, HuggingFace, NLP and speech processing
- **Web Development:** Python, C++, Golang, SQL; TypeScript, Vue3; Docker, Git, Linux
- **Languages:** English (proficient), Chinese Mandarin (first language), Japanese (daily communication)

INTERNSHIP

Tencent AI Lab, Shenzhen, China

Feb 2025 – Jul 2025

Research Intern, under the supervision of Dr. Zhifeng Li

- Researched and developed a multitask speech generation model, achieving a 16.4% relative word error rate (WER) reduction over the baseline.
- Implemented various multimodal alignment strategies to integrate text, PPG, and mel-spectrogram inputs into a conditional flow matching (CFM) model.
- Implemented the classifier-free guidance (CFG) that support multiple tasks (text-to-speech, voice conversion) and enable fine-grained control of the influence strength from different input conditions.
- Trained the model with 300M+ parameters on 16-GPU clusters and conducted quantitative evaluations.

PKU-ByteDance Open Lab for Digital Humanities, Beijing, China

Nov 2024 – Feb 2025

Software Development Intern

- Full-stack development for digital humanities research agents integrating LLMs with user-facing tools.
- Implemented an interactive map component with Vue3 frontend and Python FastAPI backend, supporting user queries of place names and visualizing results on the map.
- Designed and implemented function-calling APIs to empower the LLM to autonomously execute geospatial queries and return structured JSON for map visualization.

UMass Amherst, Amherst, MA

Oct 2023 – Oct 2024

Research Assistant, under the supervision of Prof. Chuang Gan

- Contributed to research on speech synthesis subsystems within multimodal generation frameworks.
- Collected and annotated custom datasets via large-scale web crawling.
- Trained speech synthesis models and baselines on curated datasets.
- Evaluated systems using objective metrics to benchmark performance.

PUBLICATIONS

- ICCV 2025 (co-author) – *RapVerse: Coherent Vocals and Whole-Body Motions Generations from Text.*
- ICASSP 2026 (first author) – *F5E-TTS: Enhancing Speech Synthesis by Aligning Text with Rich Semantic Representations*